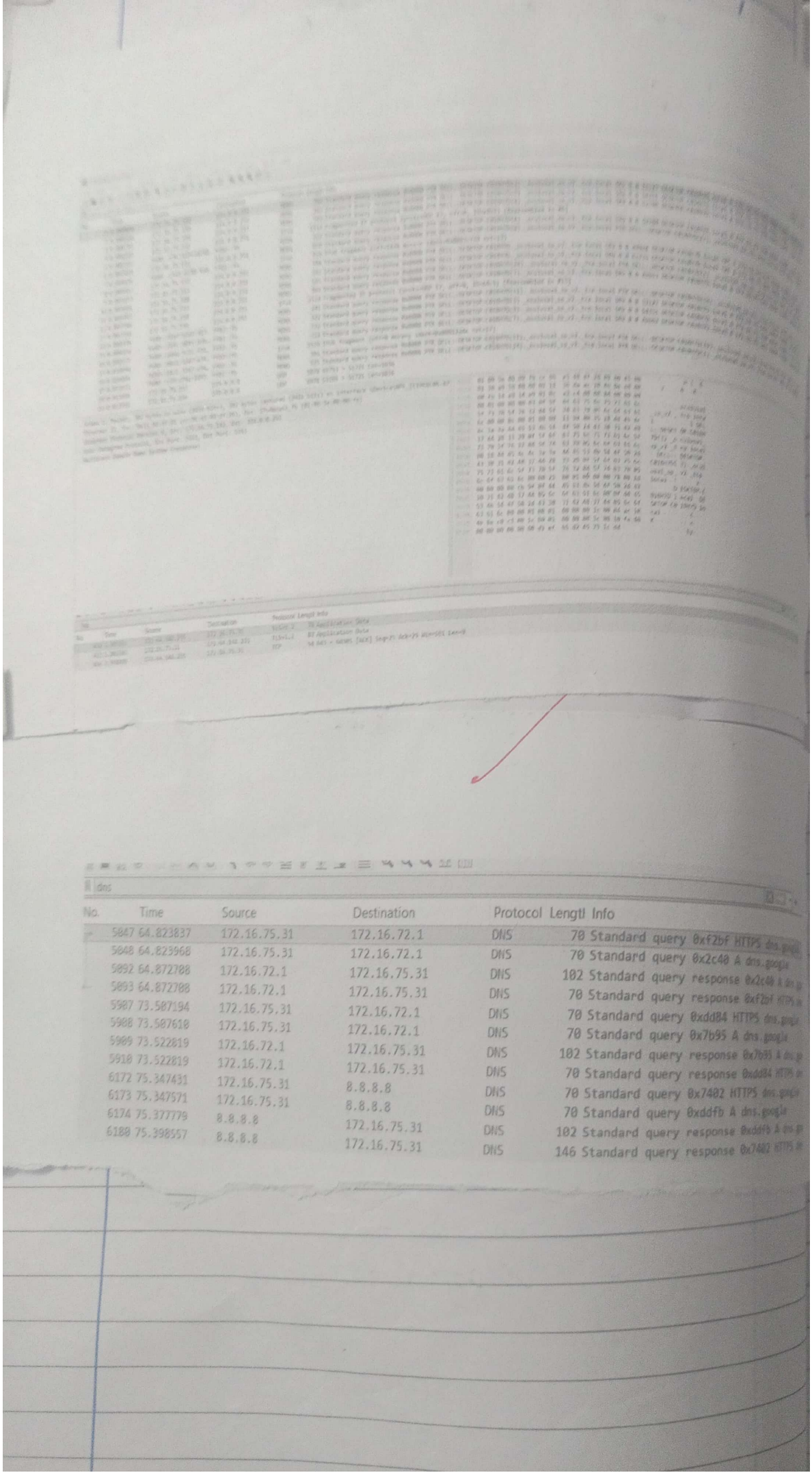


dns

No.	Time	Source	Destination	Protocol	Length	Info
5047	64.823837	172.16.75.31	172.16.72.1	DNS	70	Standard query 0xf2bf HTTPS dns.google
5048	64.823968	172.16.75.31	172.16.72.1	DNS	70	Standard query 0x2c40 A dns.google
5092	64.872788	172.16.72.1	172.16.75.31	DNS	102	Standard query response 0x2c40 A dns.google
5093	64.872788	172.16.72.1	172.16.75.31	DNS	70	Standard query response 0xf2bf HTTPS dns.google
5907	73.507194	172.16.75.31	172.16.72.1	DNS	70	Standard query 0xdd84 HTTPS dns.google
5908	73.507610	172.16.75.31	172.16.72.1	DNS	70	Standard query 0x7b95 A dns.google
5909	73.522819	172.16.72.1	172.16.75.31	DNS	102	Standard query response 0x7b95 A dns.google
5910	73.522819	172.16.72.1	172.16.75.31	DNS	70	Standard query response 0xdd84 HTTPS dns.google
6172	75.347431	172.16.75.31	8.8.8.8	DNS	70	Standard query 0x7402 HTTPS dns.google
6173	75.347571	172.16.75.31	8.8.8.8	DNS	70	Standard query 0xddfb A dns.google
6174	75.377779	8.8.8.8	172.16.75.31	DNS	102	Standard query response 0xddfb A dns.google
6180	75.398557	8.8.8.8	172.16.75.31	DNS	146	Standard query response 0x7402 HTTPS dns.google

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30/7/26 Exp 5 Exp on Packet Capture tool

Aim :- To understand the feature of Wireshark as a Packet capture tool and understand encapsulation of information at various layers of a protocal stack

Packet Sniffer :-
* Capture Packets Sent and received by the Computer
* Display protocal information of captured Packets
* Works Passively
* Receives copies of Network Packet for analysis

Packet Sniffer Tools :-
* Tcp dump - Command line Packet capture tool
* Wireshark - Graphical Packet analyzer.

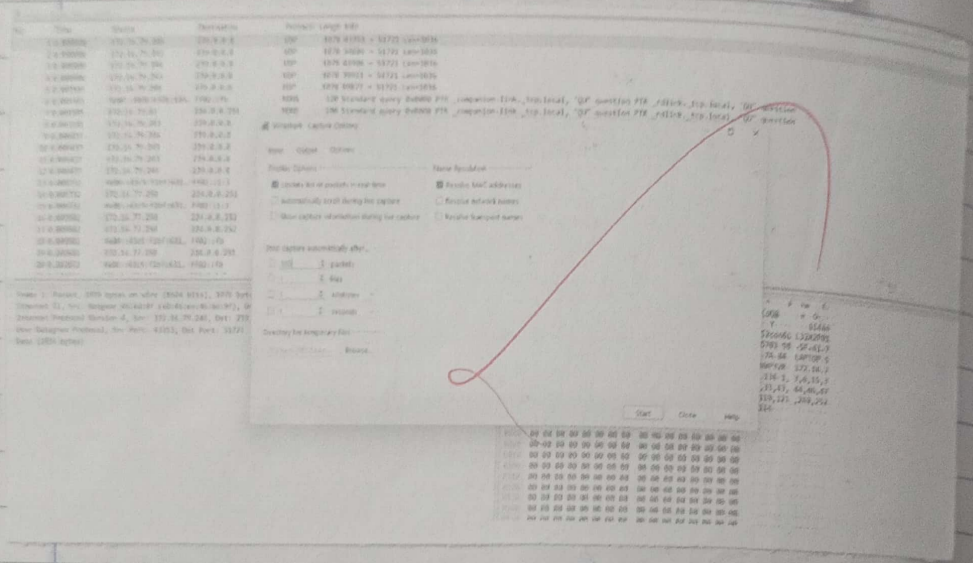
Description :-
Wireshark is a Network protocol analyzer that capture and displays Network Packets in real time. It helps monitor, analyze, troubleshoot & inspect network traffic using filters and Packet details.

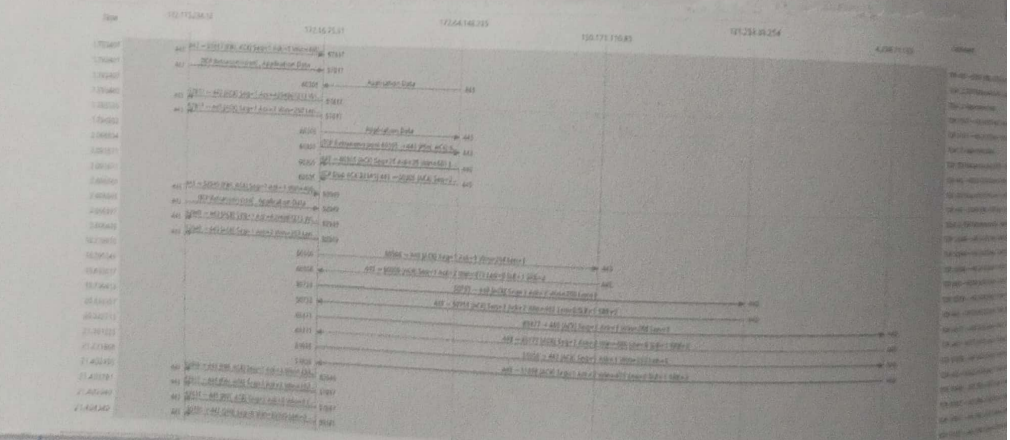
Uses of Network :-
* Capture network traffic
* Analyze Protocols.
* Applies Display filters.
* Monitors Network Issues.

DOMS

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File: Name	Filter Expression
Ethernet address 00:00:5e:00:53:00	eth.addr == 00:00:5e:00:53:00
Ethernet type 0x0806 (ARP)	eth.type == 0x0806
Ethernet broadcast	eth.addr == ff:ff:ff:ff:ff:ff
No ARP	not arp
IPv6 only	ip
IPv4 address 192.0.2.1	ip.addr == 192.0.2.1
IPv4 address isn't 192.0.2.1	ip.addr != 192.0.2.1
IPv6 only	ipv6
IPv6 address 2001:db8::1	ipv6.addr == 2001:db8::1
TCP only	tcp
UDP only	udp
Non-DNS port	!(udp.port == 53 || tcp.port == 53)
TCP or UDP port is 80 (HTTP)	tcp.port == 80 || udp.port == 80
HTTP	http
No ARP and no DNS	not arp and not dns
Non-HTTP and non-SMTP to/from 192.0.2.1	ip.addr == 192.0.2.1 and tcp.port not in {80, 25}





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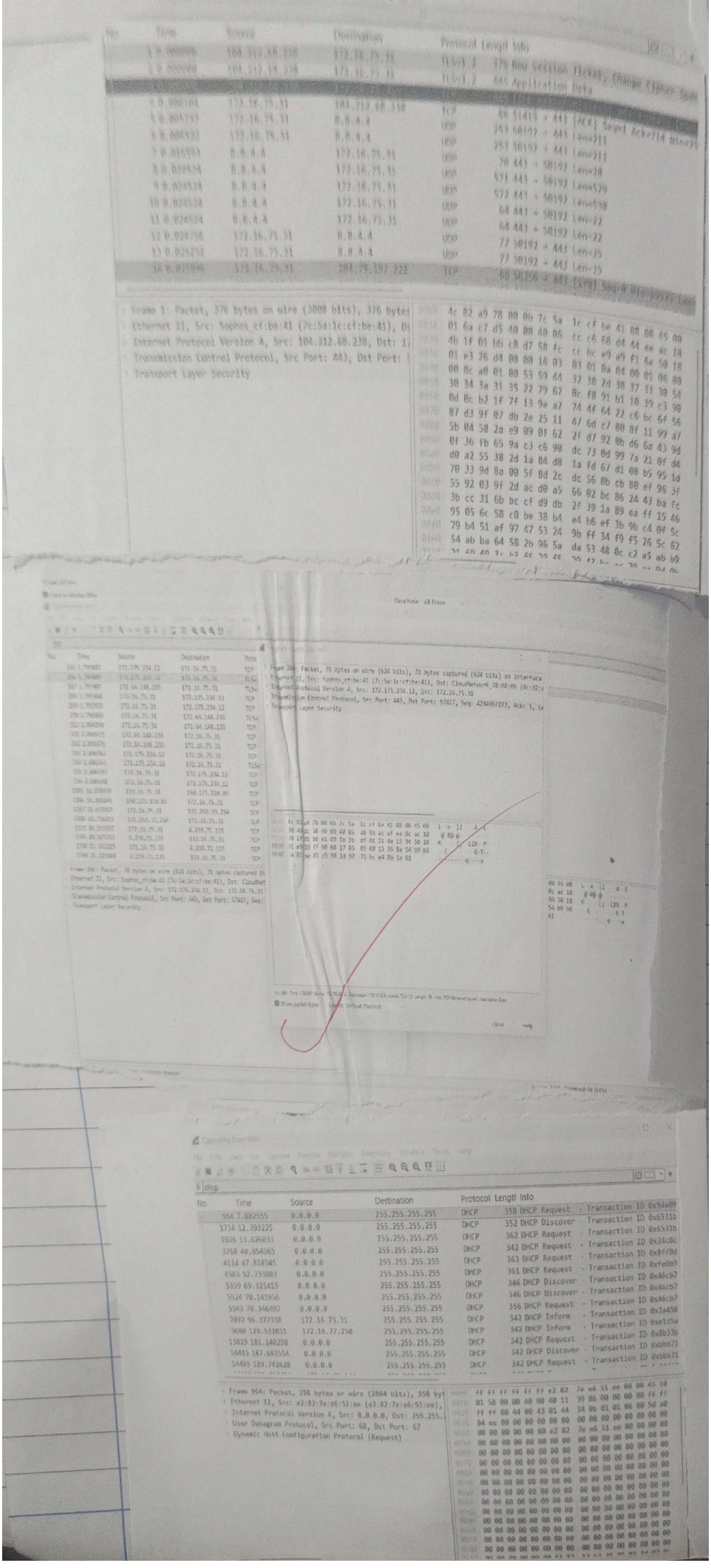
Applications:-
* Network troubleshooting
* Security analysis
* Protocol debugging
* Learning network Protocols.

Capturing Packet:-
1. Open Wireshark
2. Select the network Interface
3. Click Start capture.
4. Observe Packet in real time.
5. Click Stop and Save to the capture.

Wireshark Panes:-
* Packet List:- Display captures Packets
* Packet Details:- Shows Protocols Information
* Packet Bytes:- Display Packet data in hexadecimal points.

Color Coding:-
* Purple:- TCP
* Blue:- UDP
* Black:- Error Packets.

Filtering Packets:-
* Enter a filter in the filter bar.
* Right click to apply filters or follows the Tcp Stream.



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Flow Graph :-
= Displays the communication flow b/w devices, making Packing, exchange easier to understand.
Capture 100 Packets
1. open Wireshark
2. Select the network Interface.
3. Set the capture limit Into 100 Packets
4. Start capturing
5. Start and Save the captured Packets

Analysing Packets using Wireshark Tools:-
1. Create a filter to Displays only Tcp / UDP Packets.
   1. open the capture file
   2. Apply the tcp || udp filter.
   3. Inspect and View the flow graph.

2. Create a file to display only ARP Packets
   1. open the capture file
   2. Apply the arp filter
   3. Inspect and View the flow graph.

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3. Create a filter to display only DNS Packets

1. Open the capture file
2. Apply the DNS filter.
3. Inspect and View the flow Graph.

4. Create a filter to display only IP/ Icmp Packets

1. Open the capture file
2. Apply the Ip or Icmp filter.
3. Inspect and View the flow Graph and Save capture.

5. Create a filter to Display only DHCP Packet

1. Open the capture file
2. Apply the Dhcp file.
3. Inspect and View the flow Graph and Save the capture.

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Part B
Fill the Gaps
and the Explanation of Difficult
Words Correctly Completed.

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